Linear Regression Question:

1. **A Student records the following dataset : x = 1,2,3 and y = 2,4,6**
    - **Find the Hypothesis function.**
    - **Calculate cost function value.**

2. **The relationship between numbers number of hours studied and marks obtain is given as:**
   (1,3),(2,5),(3,7),(4,9)
    - **Compute slope and intercept.**
    - **Use y = mx+c equation**
    - **Find mean squared error cost function .**

3. **A machine produce the following output level :**
   (2,5(,(4,9),(6,13)
    - **Obtain the best finding straight line.**
    - **Write the hypothesis and y = mx+c both functions.**
    - **Calculate cost j.**

4. **Research studied effect of training hours on performance score:**
   (1,35),(2,42),(3,50),(4,63),(5,72)
    - **Find regression equation.**
    - **Predict score when hours 6.**
    - **Compute cost value.**

5. **The data observed is:**
   (1,2),(2,5),(3,5),(4,8),(5,10)
    - **Determine best fit line.**
    - **Find cost function.**
    - **Comment of goodness of fit.**

6. **Implement all question also practical coding .**